



Hemoglobin concentration in pregnancy and perinatal mortality: A London-based cohort study

Mark P. Little, DPHIL,^{a,*} Pauline Brocard, MPHIL,^b Paul Elliott, MD, PhD,^a
Philip J. Steer, MD^b

Department of Epidemiology and Public Health,^a and Academic Department of Obstetrics and Gynaecology,^b Imperial College London, Faculty of Medicine, St Mary's Campus, Norfolk Place, London, United Kingdom

Received for publication August 12, 2004; revised November 12, 2004; accepted November 23, 2004

KEY WORDS

Perinatal mortality
Stillbirth
Neonatal mortality
Hemoglobin

Objective: Failure of fetal growth during pregnancy, and preterm birth, are the major causes of stillbirth and early neonatal death. The objective of the study was to determine the association between maternal hemoglobin concentration during pregnancy and perinatal mortality.

Study design: The design was prospective, using data on 222,614 first singleton pregnancies in the St Mary's Maternity Information System database in the Northwest Thames region of London.

Results: The association of perinatal mortality with maternal hemoglobin at first antenatal check was not statistically significant ($P > .10$), but a statistically significant ($P < .001$) U-shaped pattern was found with lowest recorded maternal hemoglobin concentration. Both early neonatal mortality and stillbirth rates were statistically significantly ($P < .005$) associated with lowest maternal hemoglobin concentration. The relationship of lowest hemoglobin with early neonatal mortality was largely mediated by the effect of preterm birth, and that between lowest hemoglobin and stillbirth by fetal growth restriction. The lowest perinatal mortality was associated with a lowest recorded maternal hemoglobin concentration of between 9–11 g/dL.

Conclusion: There is an optimal range of lowest hemoglobin concentration in pregnancy, and on either side of this perinatal mortality is increased. The effect of lowest hemoglobin is largely mediated through associations with preterm birth and fetal growth restriction.

© 2005 Elsevier Inc. All rights reserved.

Failure of fetal growth during pregnancy, and preterm birth, are the major causes of stillbirth and neonatal death. A vital determinant of fetal growth is the ability of the mother to increase her cardiac output by up to 50% so as to perfuse the placenta efficiently.^{1,2} To facilitate this,

every pregnant woman retains fluid, and in the most successful pregnancies there is a 50% expansion of plasma volume. The total red cell mass also increases, but not by so much so that the hemoglobin (Hb) concentration falls during pregnancy.² Failure of the Hb concentration to fall in early pregnancy and then remain low until term is associated with a poor pregnancy outcome, with an increase in the incidence of preeclampsia, intrauterine growth restriction, preterm birth, and stillbirth.^{3–5} The reason for this remains unclear, but one proposed hypothesis is that hemodilution may occur in order to

Pauline Brocard was supported by the Save the Baby Fund.

* Reprint requests: Dr M. P. Little, Department of Epidemiology and Public Health, Imperial College Faculty of Medicine, St Mary's Campus, Norfolk Place, London W2 1PG, UK.

E-mail: mark.little@imperial.ac.uk

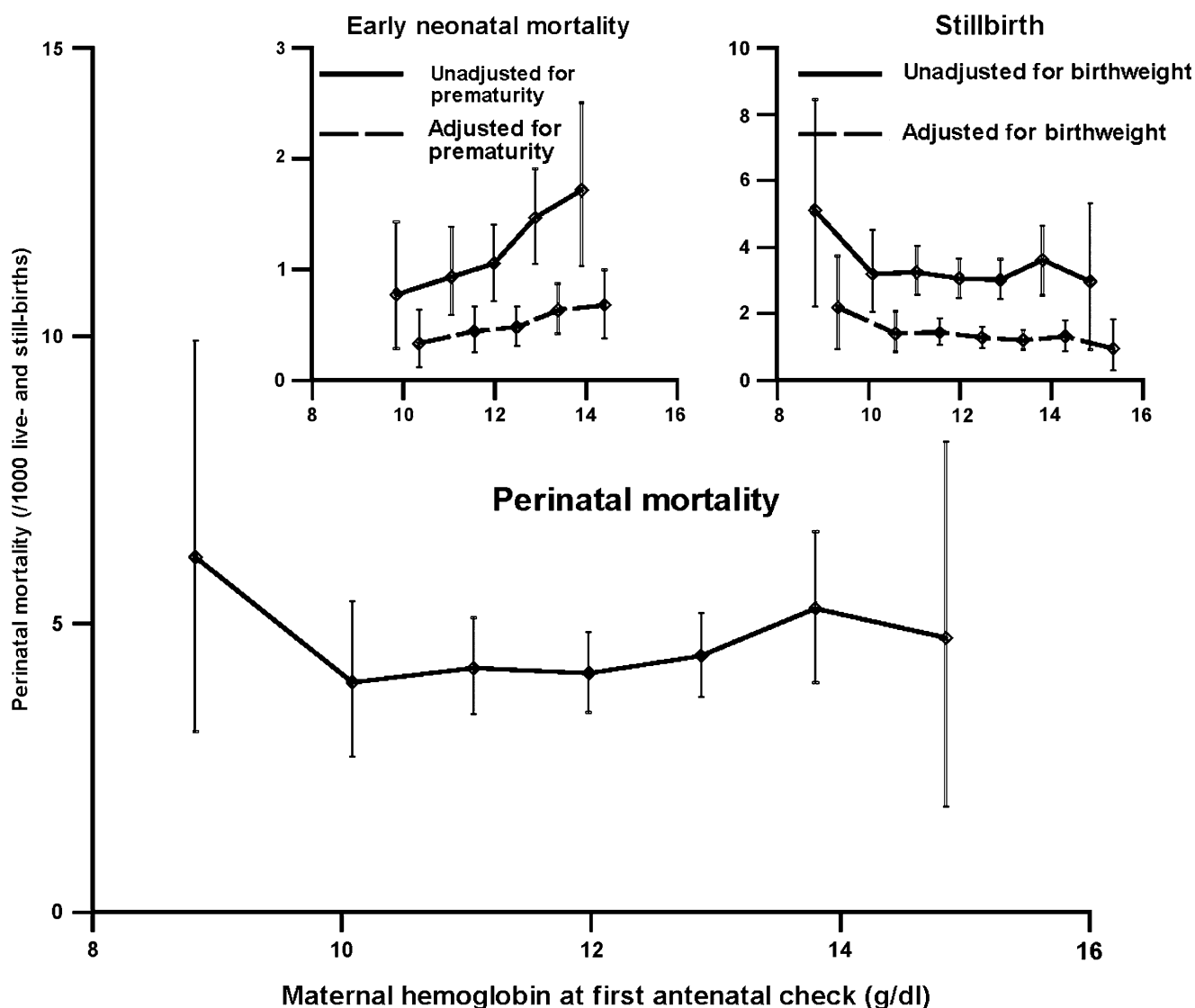


Figure 1 Perinatal and early neonatal mortality and stillbirth (with 95% CI) in relation to the maternal hemoglobin level at first antenatal check; for early neonatal mortality additionally adjusted or unadjusted for prematurity (weeks of gestation), for stillbirth additionally adjusted or unadjusted for birthweight. Adjusted for mother's ethnic group, smoking habits, maternal height, maternal weight, mother's age at first antenatal check, Carstairs' index and calendar year; all variables plus perinatal outcome, birth weight, parity, number of infants in delivery, maternal hemoglobin level at first antenatal check, and lowest maternal hemoglobin level available for 144,209 births.

reduce blood viscosity, and thus, improve placental perfusion.² In this study we analyzed one of the largest validated population-based pregnancy databases in the world, the St Mary's Maternity Information System database, to investigate the associations between the Hb concentration at first antenatal check, the lowest recorded Hb concentration during pregnancy, and perinatal mortality.

Material and methods

The St Mary's Maternity Information System (SMMIS) database contains information on birth weight, perinatal

mortality, Hb concentration at the first antenatal check, and the lowest recorded Hb concentration during pregnancy. The first antenatal check is usually at about 14 weeks' gestation. Hb level is checked at that time, and then routinely at least twice more in pregnancy, at 26 to 28 weeks', and again at 34 to 36 weeks. In most cases, the 26- to 28-week value is the lowest, with a slight rise at 34 to 36 weeks. Constraints on the amount of data that busy clinicians are willing to record into the database means that not every Hb measurement is entered; rather, the clinician (usually a midwife) is asked to record a value representing the lowest antenatal level after the first visit. The database currently contains

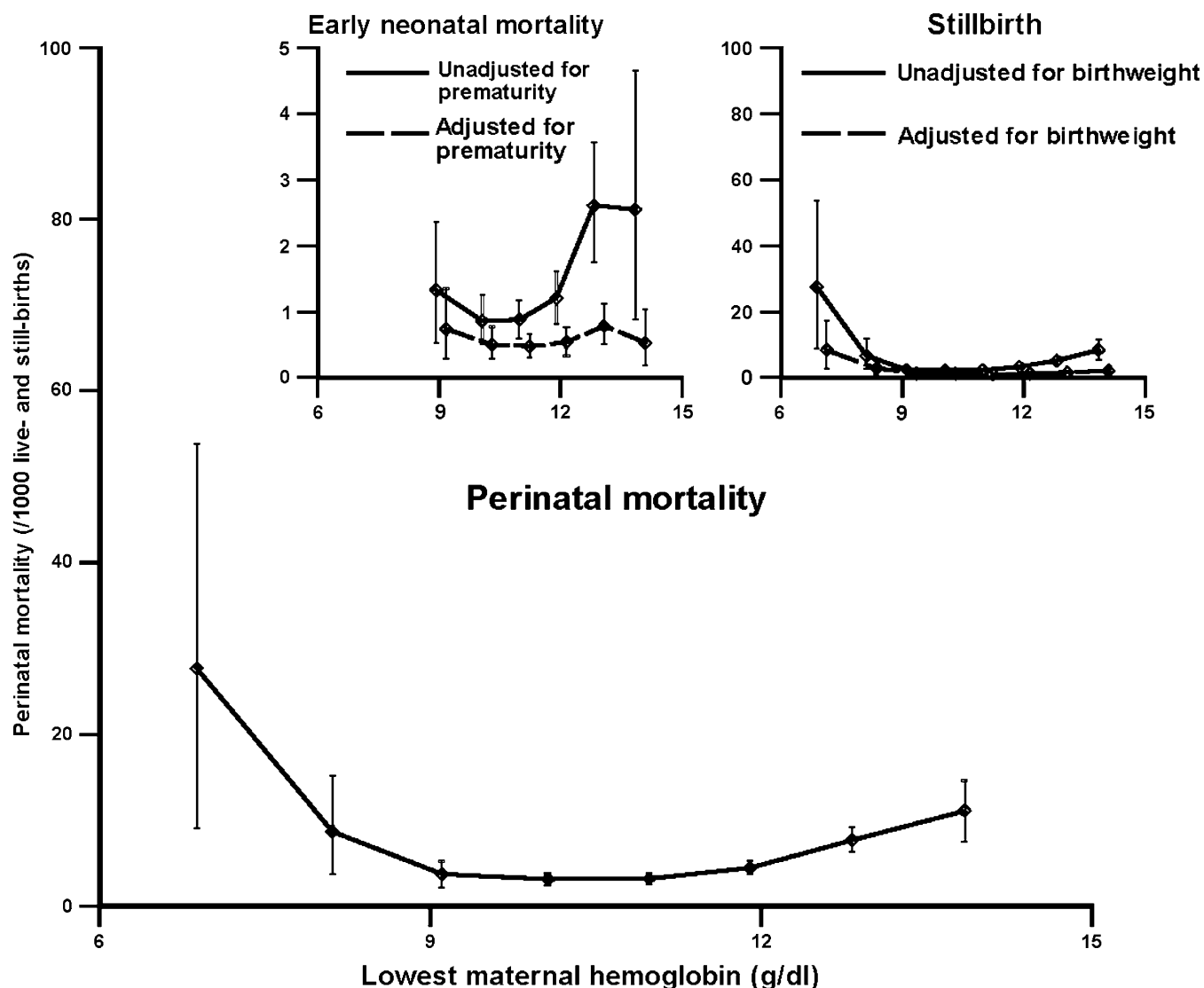


Figure 2 Perinatal and early neonatal mortality and stillbirth (with 95% CI) in relation to the lowest maternal hemoglobin level; for early neonatal mortality additionally adjusted or unadjusted for prematurity (weeks of gestation), for stillbirth additionally adjusted or unadjusted for birth weight. Adjusted for mother's ethnic group, smoking habits, maternal height, maternal weight, mother's age at first antenatal check, Carstairs' index and calendar year; all variables plus perinatal outcome, birth weight, parity, number of infants in delivery, maternal hemoglobin level at first antenatal check, and lowest maternal hemoglobin level available for 144,209 births.

585,291 pregnancies that have been recorded at 17 maternity units in the Northwest Thames region using SMMIS over the period 1988 to 2000. A parallel analysis of this database examining birth weight in relation to blood pressure has recently appeared.⁶ Of these pregnancies, 517,381 progressed far enough into their pregnancy to have a birth weight recorded. We chose to study the 222,909 singleton first live- and stillbirths to ensure no woman was studied twice and to exclude the effects of multiple pregnancies. Two hundred ninety-five pregnancies had a recorded birth weight less than 200 g, which we excluded as being implausible, leaving a cohort of 222,614 live- and

stillbirths. Of these, there were 144,209 with complete information on mother's ethnic group, smoking habits, maternal height and weight, mother's age at first antenatal check, area deprivation score (Carstairs' index, an index of social deprivation derived from the national census),⁷ calendar year, Hb measured at the first antenatal check, lowest Hb, stillbirth, and early neonatal death (deaths among live born infants within 7 days of birth). There were 903 perinatal deaths in this group, of which 689 were stillbirths and 214 were early neonatal deaths.

The key features of SMMIS are that data are entered by trained clerks or midwives with on-line validation

Table I Early neonatal death and stillbirth rates (per 1000 total births) and numbers of deaths, as a function of various explanatory variables, among first singleton births

Variable	Levels	Early neonatal death rate per 1000 births at ages 0 to 6 days (number of early neonatal deaths)		Stillbirth rate per 1000 births (number of still births)	
Mother's ethnic group	White European	1.31 (143)	$P < .001$	4.17 (455)	$P < .001$
	Black African	4.02 (15)		4.56 (17)	
	Black Caribbean	3.23 (12)		8.61 (32)	
	Indian, Pakistani, Bangladeshi	1.61 (28)		7.66 (133)	
	Oriental	1.28 (3)		3.42 (8)	
	Mediterranean	2.11 (7)		4.82 (16)	
	Other	1.29 (6)		6.00 (28)	
Smoking	Non-smoker	1.51 (179)	$P = .749$	4.66 (55)	$P = .344$
	1-9 cigarettes/d	1.20 (17)		4.88 (69)	
	10-19 cigarettes/d	1.71 (15)		5.71 (50)	
	> 20 cigarettes/d	1.20 (3)		6.38 (16)	
Maternal body mass index (BMI) (kg/m^2)	BMI < 20	1.46 (28)	$P = .151$	3.92 (75)	$P < .001$
	20 $\leq$ BMI < 25	1.33 (109)		4.38 (358)	
	BMI $\geq$ 25	1.78 (77)		5.92 (256)	
Mother's age at first antenatal check	0-24 y	1.32 (62)	$P = .297$	4.72 (221)	$P < .001$
	25-29 y	1.42 (76)		3.78 (203)	
	30-34 y	1.83 (61)		5.60 (187)	
	> 35 y	1.45 (15)		7.55 (78)	
Carstairs' deprivation quintile	1 (most affluent)	1.13 (34)	$P = .134$	4.43 (133)	$P = .376$
	2	1.48 (47)		4.69 (149)	
	3	1.55 (55)		5.00 (178)	
	4	1.40 (40)		4.45 (127)	
	5 (most deprived)	2.08 (38)		5.57 (102)	
Year	1988-1990	1.56 (57)	$P = .835$	4.58 (167)	$P = .666$
	1991-1993	1.36 (58)		4.81 (205)	
	1994-1996	1.59 (50)		5.16 (162)	
	1997-2000	1.45 (49)		4.58 (155)	

and prompting, and standardized clinical definitions. Importantly, the data are entered in near to real time and are subjected to input data quality checks. Input ranges are defined so that unlikely values are questioned. This ensures that the data are of high quality.⁸

In order to examine perinatal mortality in relation to maternal Hb while simultaneously adjusting for possibly confounding variables (mother's ethnic group, smoking habits, height, weight, age at first antenatal check, Carstairs' index, calendar year), logistic models are fitted to the data by maximum-likelihood techniques.⁹ To control for the effects of fetal growth restriction and prematurity (weeks of gestation at delivery), additional adjustments for birth weight and weeks of gestation were employed in analyses of stillbirth and early neonatal mortality, respectively. Very similar adjustments were employed in previous analysis of this data.⁶ Additional logistic regression analysis of stillbirth in relation to birth weight, and early neonatal mortality in

relation to prematurity, were also carried out, adjusted in similar ways as above. It is important to adjust both stillbirth and early neonatal death for some measure of fetal development, which is known to adversely affect both outcomes. Birth weight and prematurity are to some extent equivalent ways of doing this. However, arguably, prematurity is not a sensible way of adjusting stillbirth because death could occur at some unknown point before birth, and so for this end point birth weight was used as the adjusting variable. The analysis employed bootstrap methodology¹⁰ to assess model fit and statistical significance. The results of fitting these logistic models are given in [Figures 1 and 2](#). For the analyses of [Tables I to III](#), 9999 bootstrap samples were used, and for the more computationally demanding logistic regression analyses of [Figures 1 and 2](#), 999 bootstrap samples were used.

The study was approved by the local research ethics committee at St Mary's Hospital.

Table II Early neonatal death and stillbirth rates (per 1000 total births) and numbers of deaths, as a function of maternal hemoglobin concentration at first antenatal check, among first singleton births

Maternal hemoglobin level at first antenatal check (g/dL)	Early neonatal death rate per 1000 births at ages 0 to 6 days (number of early neonatal deaths)		Stillbirth rate per 1000 births (number of still births)	
<9.5	1.56 (2)	$P = .265$	9.38 (12)	$P = .179$
9.5-10.5	1.10 (6)		5.31 (29)	
10.5-11.5	1.24 (30)		5.04 (122)	
11.5-12.5	1.29 (71)		4.59 (252)	
12.5-13.5	1.73 (78)		4.49 (202)	
13.5-14.5	1.99 (24)		5.48 (66)	
> 14.5	2.35 (3)		4.70 (6)	

Table III Early neonatal death and stillbirth rates (per 1000 total births) and numbers of deaths, as a function of lowest maternal hemoglobin concentration, among first singleton births

Lowest maternal hemoglobin level (g/dL)	Early neonatal death rate per 1000 births at ages 0 to 6 days (number of early neonatal deaths)		Stillbirth rate per 1000 births (number of still births)	
<7.5	0.0 (0)	$P < .001$	48.78 (6)	$P < .001$
7.5-8.5	3.11 (2)		12.44 (8)	
8.5-9.5	1.85 (8)		4.16 (18)	
9.5-10.5	1.12 (25)		3.64 (81)	
10.5-11.5	1.07 (55)		3.53 (182)	
11.5-12.5	1.43 (66)		4.95 (229)	
12.5-13.5	3.04 (50)		7.91 (130)	
> 13.5	3.02 (8)		13.22 (35)	

Results

The associations of perinatal mortality with key maternal variables are shown in Table I.

Table II demonstrates that there are only weak indications ($P > .10$) of heterogeneity in neonatal mortality and stillbirth rates by maternal Hb at first antenatal check.

As shown in Table III, neonatal mortality and stillbirth rates (per 1000 births) vary significantly, with lowest hemoglobin concentration being higher at both very low and very high concentrations; the lowest perinatal mortality rates correlate with lowest hemoglobin concentrations in the range of 9 to 11 g/dL. As shown in the subanalyses in Figure 2, the effect of lowest hemoglobin concentration on adverse perinatal outcome is mediated largely through stillbirth, reflecting the much larger number of stillbirths compared with early neonatal deaths in this dataset.

The logistic regression analysis given in Figure 1 to some extent confirms the results of the univariate analysis of Table II, and shows little relationship between perinatal mortality, also early neonatal mortality and stillbirth, and first antenatal check Hb. Linear and linear-quadratic function of first antenatal check Hb concentration (centered at 12 g/dL) fitted to the total

perinatal mortality data had linear and quadratic coefficients that were not statistically significantly ($P > .10$) different from zero, although the quadratic coefficient was positive at borderline levels of statistic significance ($P = .074$); there is no evidence for trends in stillbirth ($P > .10$) in relation to booking Hb (results not illustrated). However, there is a statistically significant ($P < .01$) increasing linear trend of early neonatal mortality with booking Hb, which remains statistically significant ($P < .01$) if a quadratic adjustment term is added. As shown by Figure 1, this increasing trend is substantially diminished by adjustment for prematurity, although it remains statistically significant ($P < .05$, with or without quadratic adjustment).

In contrast, the logistic regression analyses reported in Figure 2 confirm the results of the univariate analysis of Table III, and show that the relationships of perinatal mortality, and both early neonatal mortality and stillbirth, with lowest Hb in pregnancy are U-shaped, with the lowest mortality rates being associated with a lowest Hb of 9 to 11 g/dL. As expected, all mortality rates increased steadily and significantly above 11 g/dL Hb (Figure 2). However, there was also a striking and statistically significant rise in perinatal mortality and stillbirth below 9 g/dL. A linear-quadratic function of lowest Hb (centered at 11 g/dL) fitted to this data had linear and

quadratic coefficients that were, for stillbirth and perinatal mortality, statistically significantly ($P < .005$) positive; for early neonatal mortality, the linear coefficients, whether resulting from a fit of a linear or a linear-quadratic function of lowest Hb, were statistically significantly positive ($P < .005$), although the quadratic term was positive only at borderline levels of statistical significance ($P = .051$) (results not illustrated). **Figure 2** also shows that after adjustment for prematurity, the relationship of early neonatal mortality with lowest Hb largely disappeared, and linear and quadratic trend terms were no longer statistically significant ($P > .10$). Likewise, after adjustment for birth weight, the relationship of stillbirth with lowest Hb was substantially diminished, and the linear coefficient was no longer statistically significant ($P > .05$, with or without quadratic adjustment), although the quadratic trend term remained statistically significant ($P < .01$).

There is a highly statistically significant ($P < .005$) lower stillbirth rate with increasing birth weight, and there is also a highly statistically significant ($P < .005$) lower early neonatal mortality rate with increasing weeks of gestation at delivery (results not shown).

Comment

Our results show that both low as well as high levels of lowest Hb concentration during pregnancy are associated with increased perinatal mortality rates, early neonatal mortality, as well as stillbirths. The effect of lowest Hb is largely mediated through associations with preterm birth and fetal growth restriction. The correlation of the first antenatal check Hb with perinatal mortality is not statistically significant.

The finding that the effect of lowest Hb on perinatal mortality is mediated through associations with preterm birth and fetal growth restriction is unsurprising. Huisman and Arnaudse³ documented significantly higher second-trimester Hb among nulliparous women with intrauterine growth restriction. Murphy et al⁴ also found lower birth weight and increased rates of preterm delivery among women with higher Hb. Lu et al¹¹ and Blankson et al¹² documented increased odds ratios for fetal growth retardation and preterm delivery among women with elevated hematocrits. Preterm birth and fetal growth restriction are well known to be associated with increased rates of stillbirth and neonatal mortality.^{13,14}

The database analyzed is one of the largest validated population-based pregnancy databases in the world. The risk factors already known to be associated with perinatal mortality, notably mother's ethnic group, smoking habits, height, weight, age at first antenatal check, Carstairs' index, and temporal trend, were accounted for in the analyses. There is a wide diversity of ethnic groups in the population studied; the pattern

of association was the same in each. Therefore, the results can probably be generalized to the different ethnic groups, although the small numbers in some ethnic groups (**Table I**) to some extent qualifies the general applicability of our findings.

A potential limitation of the study is the substantial amount of missing data. Of the 222,614 first singleton births, there were 144,209 (64.8%) with complete information on mother's ethnic group, smoking habits, maternal height and weight, mother's age at first antenatal check, area deprivation score, calendar year, Hb measurements, and perinatal death. However, unadjusted analysis in the complete dataset, with information on perinatal death and Hb measures, as well as the other adjusting variables, yields similar findings to that of the dataset with missing information in the adjusting variables, and to the adjusted analyses given here (results not shown).

It is possible that a certain proportion of gravidas with significant anemia may have a greater number of comorbid conditions that may directly contribute to perinatal mortality (rather than via anemia), such as vascular insufficiency (eg, renal disease, systemic lupus erythematosus). It is also possible that hemoglobinopathies associated with ethnic subgroups, such as thalassemia, may account for some of the associations noted here. However, the number of such cases is likely to be very small; there are only 686 mothers with sickle cell trait, 27 with thalassemia, and 965 mothers with a history of renal disease in the cohort of 222,614 first singleton births. Given this, such factors are unlikely to materially contribute to the trends observed here. Additional analyses in which patients with thalassemia or sickle cell trait, or with a history of renal disease, yielded very similar findings (results not shown).

Our study suggests that routine iron supplementation of women with a lowest Hb of 9.0 g/dL or more is unlikely to improve the perinatal mortality rate, although maternal benefits may accrue from such intervention. A meta-analysis of 20 randomized controlled trials of iron supplementation did not show any significant benefits to this approach.¹⁵ Indeed, a more recent study has suggested that supplementing iron to women who already have good iron stores can produce an increase in preterm births,¹⁶ consistent with our findings. The importance of a fall in Hb during pregnancy, indicating a healthy plasma volume expansion, has been appreciated for some time.⁴ Nonetheless, the fact that birth weight is highest,² and perinatal mortality lowest when the Hb falls to between 9 and 11 g/dL is often a surprise to those used to the original WHO definition of anemia in pregnancy, a Hb concentration <11 g/dL.¹⁷ In our generally well-nourished population, such values more likely indicate a good response to pregnancy than true iron deficiency anemia. The fact that the first antenatal check Hb is not informative about perinatal mortality suggests that plasma

volume expansion is much more important for pregnancy outcome than the level of iron stores (the same might not be true of infant development). In our population, routine screening for first-trimester anemia and iron supplementation are unlikely to be cost effective. In the developing world, severe iron deficiency coupled with failure of plasma volume expansion might give similar lowest Hb but with a poor outcome, so our conclusions should not be extended to their populations without further studying how we could best distinguish which of these 2 processes predominates in any particular individual.

Acknowledgments

We thank the 2 referees for their detailed and helpful comments. We also thank the units that contribute data to SMMIS, the consultants in participating hospitals, and the midwives who enter the data, without whom such analyses would not be possible.

References

1. Hytten FE, Paintin DB. Increase in plasma volume during normal pregnancy. *J Obstet Gynaecol* 1963;70:402-7.
2. Steer PJ. Maternal hemoglobin concentration and birth weight. *Am J Clin Nutr* 2000;71(5 Suppl):1285S-7S.
3. Huisman A, Aarnoudse JG. Increased 2nd trimester hemoglobin concentration in pregnancies later complicated by hypertension and growth retardation. Early evidence of a reduced plasma volume. *Acta Obstet Gynecol Scand* 1986;65:605-8.
4. Murphy JF, O'Riordan J, Newcombe RG, Coles EC, Pearson JF. Relation of haemoglobin levels in first and second trimesters to outcome of pregnancy. *Lancet* 1986;i:992-5.
5. Stephansson O, Dickman PW, Johansson A, Cnattingius S. Maternal hemoglobin concentration during pregnancy and risk of stillbirth. *JAMA* 2000;284:2611-7.
6. Steer PJ, Little MP, Kold-Jensen T, Chapple J, Elliott P. Maternal blood pressure in pregnancy, birth weight, and perinatal mortality in first births: prospective study. *BMJ* 2004;329:1312-4.
7. Carstairs V, Morris R. Deprivation and health in Scotland. Aberdeen: Aberdeen University Press; 1991.
8. Cleary R, Beard RW, Coles J, Devlin HB, Hopkins A, Roberts S, Schumacher D, et al. The quality of routinely collected maternity data. *BJOG* 1994;101:1042-7.
9. McCullagh P, Nelder JA. Generalized linear models, 2nd ed. London: Chapman and Hall; 1989.
10. Davison AC, Hinkley DV. Bootstrap methods and their application. Cambridge: Cambridge University Press; 1997.
11. Lu ZM, Goldenberg RL, Cliver SP, Cutter G, Blankson M. The relationship between maternal hematocrit and pregnancy outcome. *Obstet Gynecol* 1991;77:190-4.
12. Blankson ML, Goldenberg RL, Cutter G, Cliver SP. The relationship between maternal hematocrit and pregnancy outcome: black-white differences. *J Natl Med Assoc* 1993;85:130-4.
13. Berkowitz GS, Papiernik E. Epidemiology of preterm birth. *Epidemiol Rev* 1993;15:414-43.
14. Cnattingius S, Haglund B, Kramer MS. Differences in late fetal death rates in association with determinants of small for gestational age fetuses: population based cohort study. *BMJ* 1998;316:1483-7.
15. Mahomed K. Iron supplementation in pregnancy. *Cochrane Database Syst Rev* 2004; (3):CD001135. Available at: <http://www.cochrane.org/cochrane/revabstr/AB000117.htm>. Accessed May 5, 2004.
16. Lao TT, Tam KF, Chan LY. Third trimester iron status and pregnancy outcome in non-anaemic women; pregnancy unfavourably affected by maternal iron excess. *Hum Reprod* 2000;15:1843-8.
17. WHO Expert Committee. Nutrition in pregnancy and lactation; report of a WHO Expert Committee. *World Health Org Techn Rep Ser* 1965;302:5-19.